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AES-R9-2008

STANDARDS

AES standards project report - Considerations for standardising AES metadata sets

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AES standards project report - Considerations for standardising AES metadata sets

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Abstract
Digital management of audio assets - in projects, inventories, collections, libraries, and archives - will increasingly be controlled by metadata; that is, data that describes the audio in some way. Some audio-related metadata already exists in metadata sets developed by other communities, however there is a need within the audio community for more specific and detailed audio metadata.

The question was raised concerning how the upcoming AES metadata standards might be handled in a way that would be consistent across a number of AES documents, and also capable of being harmonised with relevant external metadata sets such as the SMPTE metadata dictionary standards (RP210 metadata elements, RP2009 metadata groups etc.) and the IEC work in the same field.

This report discusses primary issues and outlines some elementary considerations.

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Foreword

This foreword is not part of the *AES standards project report - Considerations for standardising AES metadata sets*, AES-R9-2008.

The question was raised at the SC-03-06 and SC-03-07 meetings in Vienna, 2007-05, (and previously) how the upcoming AES metadata standards might be handled in a way that was consistent across the many AES documents, and also capable of being harmonised with the SMPTE metadata dictionary standards (RP210 metadata elements, RP2009 metadata groups etc.) and the IEC work in the same field.

In 2007-07, an ad-hoc task group met to consider some questions that are important for our standardisation of metadata in general. During subsequent discussions in the New York meeting of SC-03-07, 2007-10, the working group felt that this report - originally intended as an internal summary - would be a useful as a stimulus to wider discussion.

The members of the writing group that developed this document in draft are: C. Chambers, M. Yonge, and D. Ackerman.

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Chair, working group SC-03-07

AES standards project report - Considerations for standardising AES metadata sets

Introduction

Digital management of audio assets - in projects, inventories, collections, libraries, and archives - will increasingly be controlled by metadata; that is, data that describes the audio in some way. Some audio-related metadata already exists in metadata sets developed by other communities, however there is a need within the audio community for more specific and detailed audio metadata.

The question was raised concerning how the upcoming AES metadata standards might be handled in a way that would be consistent across a number of AES documents, and also capable of being harmonised with relevant external metadata sets such as the SMPTE metadata dictionary standards (RP210 metadata elements, RP2009 metadata groups etc.) and the IEC work in the same field.

This report discusses primary issues and outlines some elementary considerations.

1. What assumptions do we start with?

Metadata is organised in sets

A parameter standing on its own has little meaning. A set of parameters, on the other hand, provides an overall context that ensures consistent meaning for all parameters in that set.

Metadata sets benefit from standardisation

Anyone can create a set of metadata and use it satisfactorily within a closed application. Where metadata needs to be handled in several applications developed and used by different companies and organisations, there needs to be a common understanding of what the metadata means.

A set of metadata developed and published as an open standard provides a common basis for meaningful metadata exchange for all parties.

Multiple metadata sets will exist

Audio is not a discrete discipline. It overlaps many others. Any parameter may be described in more than one metadata set and, potentially, from more than one standards body. In the general case, practical applications will need to handle multiple metadata sets in a predictable way.

The meaning of any parameter may be distorted if it is not considered as part of a defining set. Any handling technique must know what set the parameter is related to.

An application may use more than one metadata set

If an application requires metadata to describe picture and sound, it is as unreasonable to expect the picture specialists to completely describe sound as it would be for the sound specialists to completely describe pictures.

Even within a nominally single discipline there will be meaningful subdivisions of expertise. It would be as unreasonable for sound archivists to be responsible for metadata for current sound production as vice versa.

Accordingly, an application must legitimately be able to refer to more than one metadata set.

Metadata sets will be incomplete

No living and growing metadata set can ever be complete. In any real-world application, there may be things to describe that are not part of a standard set. There needs to be a method to handle such new items without breaking the standard sets, or the metadata scheme itself. These new items may, of course, be adopted in a future revision of a standard set.

Metadata transport schemes will change

Schemes for coding and transporting metadata may continue to change over time. A good metadata set will not be tied to a single transport scheme, but will continue to be useful with alternative transports and coding schemes in the future.

Metadata schemes of current interest use Key-Length-Value (KLV) coding or Extensible Markup Language (XML). Either notation should be adequate to define an element unambiguously. An element described in XML should be capable of mapping transparently to and from KLV.

2. What questions arise?

2.1 How to identify each metadata set?

Whatever transport or coding scheme is used to convey metadata, and for each piece of metadata, there needs to be a way to identify the set that unambiguously defines the meaning of that metadata.

XML namespaces. XML Namespaces (for example) will identify which metadata set defines the metadata elements and attributes. See W3C info on namespaces. http://www.w3.org/TR/1999/REC-xml-names-19990114/

KLV unique group labels. SMPTE KLV 336M group coding include an identifying unique label that identifies the metadata set.

2.2 Managing set overlap

2.2.1 How can we inter-relate different metadata sets?

Some sets may, by design, define overlapping parameters consistently, so that mapping between them will be completely transparent and lossless.

Where transparency cannot be assured, a mapping process translates metadata values from one set to another, or to a consistent format internal to the application.

2.2.2 Can we identify compatible metadata sets to cover the whole spectrum of potential contributors and users?

Metadata sets from specialist interest groups become more valuable where they can demonstrate compatibility and can be used together cooperatively to cover a wider range than any single set.

2.3 How can we map metadata between different sets?

Where transparency cannot be assured, there is a risk of mapping loss; metadata values converted from one set to another may not be capable of being converted back again to be identical with the original value. The usefulness of the metadata transferred in such a manner becomes compromised.

Clear type definitions help to make a mapping outcome predictable. For example, if a pair of definitions from different sets unambiguously state that both values represent the “clip name or title”, accurate mapping is reasonably certain. However, if one value was defined as “name” (associated with one set of conventions) while the other was defined as “title” (associated with a different set of conventions), then the mapping between the two values may be ambiguous and could contain a value that represents a number of other possibilities.

Who will be responsible for the mapping? Unless there is clear understanding of where the data will be used, the exporting system will not be able to predict the capabilities of the importing system but should export a clearly-identified metadata set. The importing system will be able to see the metadata-set declaration and should map accordingly if its internal metadata assumptions are different. The onus for mapping therefore falls on the receiving/importing system.

3 Metadata standards criteria

3.1 Metadata quality

What separates good metadata from poor metadata? This isn’t often discussed and we believe it’s fundamental for long-term usefulness of metadata sets. There will be plenty of choice for metadata sets; the ones that succeed in the longer term will be the ones that provide qualities that the ad-hoc schemes will lack, for example:

- Clarity
- Accuracy
- Precision
- Completeness
- Lack of ambiguity
- Scaleability
- Mapability

A guide to insuring that consistent quality is maintained can be found in ISO 11179: *Information Technology, Metadata Registries.*

3.2 Metadata maintainability

Each standard set needs human-readable definitions to create, agree, and maintain the elements easily.

The definitions used at the point of exchange should also be machine-readable.

4 AES metadata standards

4.1 Registry

AES should maintain its own registry to provide an authoritative reference for AES metadata. This can be registered as a Class 13 registry within the SMPTE MDD.

4.2 Metadata sets for archives

Metadata sets for audio archives have already been identified. Active projects include:

AES-X098A, Descriptive metadata for audio objects—Core audio schema
AES-X098B, Administrative and structural metadata for audio objects
AES-X098C, Administrative metadata for audio objects—Process history schema
AES-X134, Core metadata for audio exchange

4.3 Minimum metadata set for audio interchange

It may be useful to assemble an authoritative set of audio metadata for current production and distribution. Part of this could be developed from the current AES Technical Document on "Recommendation for delivery of recorded music projects".

5 Action

5.1 Seek cooperative candidate metadata sets

The case for compatible inter-operation with other metadata sets has been made. There will be pressure for each audio community to create their own metadata set (because they can easily start such a task - finishing it is harder, of course). We need to identify existing metadata sets and examine how their metadata content could overlap and with AES sets.

We felt that the most-developed template was the SMPTE Metadata Dictionary (see SMPTE 335M and the audio components within RP210).

4.2 Build from a strong core

It will be preferable to develop from a common core and add specialist sets only where necessary. Overlap between specialist sets needs to be managed carefully - this is where the greatest degree of incompatibility will be found.

Summary

Keep metadata standards separate from any implementation

- Good metadata should be defined once, but be capable of implementation across different platforms and transport schemes. This requires a consistent approach. (See ISO 11179).
- AES metadata should be consistent across AES standards.
- AES metadata should map consistently to SMPTE metadata
- Simplified forms should be simplified in predictable ways.
- Mapping of the same metadata between different implementations (XML, KLV, etc.) should be transparent.
- Structures should be extensible in a layered (or trunk and branch) structure such that core sets can form a part of common standards and specialist additions be added in a consistent and identifiable way.

Adopt a standard text presentation

- Name
- Unique Tag (XML)/ Unique Label (SMPTE)
- Model structure (level in hierarchy)
- Short definition
- Full Definition, or reference
- Data type
- Value length

AES internal metadata policy

- Form a top-level AES registry node; establish a AES clear namespace.
- Relate AES metadata to cooperative disciplines
- Associate AES top-level node with SMPTE MDD Class 13 registration

Coordinate different AES metadata topics

- Working-group responsibility for subnodes
- Subcommittee responsible for node coordination

Consider implementation guidelines for metadata

For example, in an implementation, when should metadata be:

1. Required
2. Permitted
3. Deprecated
4. Illegal